

REES ALGEBRAS AND BLOWUP ALGEBRAS

PRISHITA DHARAMPAL

(1) **TODO: Motivation**

(2) Homogeneous prime ideals encode homogeneous polynomials vanishing on some projective subvariety of the ring. For instance, given the (homogeneous coordinate) ring $k[x_0, \dots, x_n]$, in its associated projective space is $P^n = \text{Proj}(k[x_0, \dots, x_n])$ we have

- (0) — corresponds to all of $\mathbb{P}^n$
- (x_i) — a hyperplane $\{x_i = 0\} \cong P^{n-1}$
- (x_i, x_j) — a codimension-2 linear subspace $\{x_i, x_j = 0\} \cong P^{n-2}$
- (f) for homogeneous irreducible f of degree d - a hypersurface $\cong P^{n-d}$.

(3) We want homogeneous primes because in $\mathbb{P}^n$ coordinates are only defined upto scaling. So a polynomial f only vanishes consistently at a projective point if $f(\lambda x) = 0$ whenever $f(x) = 0$, which forces f to be homogeneous.

(4) Graded Rings.

- Worth noting that all ring maps from $R \rightarrow \text{gr}_I R$ map through the quotient R/I .
- The blow up algebra surjects onto $\text{gr}_I R$,

$$B_I R / I B_I R \cong \text{gr}_I R$$

- If $\text{gr}_I R$ is a domain, then R is a domain (converse not true).

(5) Let M be a smooth manifold and k be a field. $C^\infty(M)$ is the ring of infinitely differentiable (smooth) functions $f : M \rightarrow k$.

(6) Let X be a topological space, $x \in X$ a point, and maps $f, g : X \rightarrow k$, then f and g are equivalent if they agree on some open neighborhood U of x such that

$$f \sim_x g \iff f|_U(u) = g|_U(u), \forall u \in U$$

The germ of f at x is its equivalence class $[f]_x$.

(7) The ring of germs at x is,

$$\mathcal{O}_{X,x} = \{[f]_x \mid f : U \rightarrow k\}$$

with addition and multiplication defined representativewise. This is a local ring with maximal ideal

$$\mathfrak{m}_x = \{[f]_x \in \mathcal{O}_{X,x} \mid f(x) = 0\}$$

which is maximal because evaluation at x gives a surjective ring homomorphism:

$$\text{ev}_x : \mathcal{O}_{X,x} \twoheadrightarrow k, \quad [f]_x \mapsto f(x)$$

with kernel $\mathfrak{m}_x$, so $\mathcal{O}_{X,x} / \mathfrak{m}_x \cong k$, which is a field.

(8)

Theorem 0.1 (Krull's Intersection Theorem). *Let R be a Noetherian ring, $I \subseteq R$ an ideal, and M a finitely-generated R -module. There exists an element $r \in I$ such that*

$$(1 - r) \bigcap_{k=1}^{\infty} I^k M = 0$$

Corollary 0.1.1. *If $(R, \mathfrak{m})$ is a local Noetherian ring, $I \subsetneq R$ then*

$$\bigcap_{k=1}^{\infty} I^k = 0$$

- (9) **TODO: Graded Rings/Modules**
- (10) **TODO: Filtrations, Stable Filtrations, Initial forms, Artin-Rees**
- (11) **TODO: Blow-up Algebras, Blow Ups, Exceptional Set**
- (12) Limiting point of a line:

A line through 0 in $\mathbb{A}^r$ is determined by its direction (which corresponds to a point in $\mathbb{P}^{r-1}$). More precisely, the line through 0 and $p = (p_1, \dots, p_r)$ corresponds to:

$$[p_1 : \dots : p_r] \in \mathbb{P}^{r-1}$$

So a sequence of secant lines ℓ_{p_n} is really a sequence of points $[p_n] \in \mathbb{P}^{r-1}$. A **limiting position** is a limit point of this sequence:

$$\ell = \lim_{n \rightarrow \infty} [p_n] \in \mathbb{P}^{r-1}$$

This is well-defined because $\mathbb{P}^{r-1}$ is compact (over $\mathbb{C}$), so every sequence has a convergent subsequence.

- (13) The correspondence between the exceptional set in the blow up and the associated graded ring gives us a beautiful geometric consequence. Let $R = k[x_1, \dots, x_r]/J$ and $I = (x_1, \dots, x_r)$ where k is algebraically closed and $J \subset I$, also let $X = \mathbb{V}(J) \subset \mathbb{A}^r$ so $0 \in X$.

– basically X is the algebraic subset of $\mathbb{A}^r$ cut out by J .

Then we define the tangent cone of X at 0 as the cone composed of all lines that are the limiting positions of secant lines to X passing through the point 0. Alternatively, the tangent cone at 0 is then:

$$TC_0(X) = \overline{\{[p] \in \mathbb{P}^{r-1} : p \in X \setminus \{0\}\}} \subset \mathbb{P}^{r-1}$$

the closure of the set of directions of points on X , together with the actual lines they represent in $\mathbb{A}^r$. The closure is what captures the limiting directions — directions you can approach but which may not themselves be realized by points of $X \setminus \{0\}$. One can show that the ideal $\text{in}_I(J) \subset k[x_1, \dots, x_r]$ defines the tangent cone so that the coordinate ring of the tangent cone is $\text{gr}_{(x_1, \dots, x_r)} R$. **TODO: exercise 5.3 eisenbud**

- The tangent cone at a smooth point is the tangent line.
-

- (14) **TODO: Examples**

- (15) What makes a good family? Flatness!

- Algebraically, this means that when the family is represented by a R -algebra S , then S is flat as a R -module.

- Geometrically this means that when the family is represented by a morphism $\phi : X \rightarrow B$, then for each point $x \in X$ there is an affine neighborhood U of x and an affine neighborhood of $\phi(x)$ such that ϕ restricts to a map $U \rightarrow V$ and the corresponding map of coordinate rings $K[V] \rightarrow K[U]$ makes $K[U]$ into a flat $K[V]$ -module.
- (16) Free resolution helps "standardize" the sequence so tor can give us a measure of the failure of exactness **TODO: examples and definitions**
- (17) **TODO: Criteria for Flatness.**
- (18) **TODO: Rees Algebras as a flat family.**